VOLUME TRANSFER AND COMPONENT CUMULANTS FOR LATTICE QED IN THREE DIMENSIONS

Abstract. We transfer normalized reference-volume response estimates for three-dimensional chamber lattice QED to products of local Wick polynomials on the prescribed family of arbitrarily large tori. A two-volume partition comparison controls the normalization. Its photon contribution loses an explicit, small power of the charge; Gaussian domination controls the nonzero flux sectors. We pass the resulting product bounds to any prescribed volume-first continuum law and then to its translation-ergodic components. Under the stated reference estimates and component Gaussian bound, a probe with positive free fourth coefficient has a positive fourth cumulant in almost every component at sufficiently small charge.

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1. Introduction

Let $\mathbb{P}^{\mathrm{af}}_{n;q,\boldsymbol{\mu}}$ be the all-flux chamber QED law on $C(n)=\mathbb{Z}^3/n\mathbb{Z}^3$, with charge $q>0$, real lattice mass $\boldsymbol{\mu}$, Wilson parameter $0<\rho<1$, weights $(6,6\sqrt{2},6\sqrt{2},6)$ and antiperiodic fermions. Its real curvature is $F^{\mathrm{u}}=dA+(2\pi/q)z$, where z is a closed integer two-cochain. Write Z_n for its unnormalized partition function and $F^{\sharp}$ for the physical step field: on the cube of side η centered at a plaquette, its corresponding component is $\eta^{-3/2}F^{\mathrm{u}}$. The measure and its normalization are recalled in section 2.

Fix an odd integer $L\geq 13$, $M=L^k$ with integer $k\geq 2$, $0<\varepsilon\leq 1/200$ and $0<\bar{m}\leq 1$. Set

$$h=L^{\lfloor \log_L(1/\bar{m})\rfloor},\qquad \eta=L^{-N},\quad q=e\sqrt{\eta},\qquad R=8hL^T,\quad n=R/\eta. \tag{1}$$

At each e,N let $\boldsymbol{\mu}_N$ be real and independent of the outer volume T. Every reference and outer law below uses this same mass. The cutoffs N,T are nonnegative integers. For $e\sqrt{h}<1$, put

$$v_K=-\log(e\sqrt{h}),\qquad r_K=\min\{L^i : i\in\mathbb{Z}_{\geq 0},\ L^i\geq v_K^2\},\qquad R^*=\frac{hMr_K}{L}=hL^{T^*}.$$

The reference torus has side $n_*=R^*/\eta$ and $N_*=8L^3$ cells, each of physical width $\ell_c=R^*/(2L)$.

Fix real nonzero $\beta\in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ and $f=dd^*\beta$. Let D_β be the side of a least coordinate cube containing $\operatorname{supp}\beta$ and let Q_β, of side $D_\beta^{\mathrm{buf}}=D_\beta+4$ and center z_β, be that cube enlarged by two in each direction. Denote cube averaging by J_η and centered differences by

$$\delta_{\eta,\mu}g=\frac{g(\,\cdot+\eta e_\mu/2)-g(\,\cdot-\eta e_\mu/2)}{\eta}.$$

Define

$$a'_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu}\beta_{\mu\nu}, \qquad f^{\eta}_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}. \tag{2}$$

The covering bond source is $\widehat{a}^{o}_{\nu}(x) = \eta^{1/2}(J_\eta a'_{\eta,\nu})(\eta x + \eta e_\nu/2)$. Put $v_\eta = \|d\widehat{a}^{o}\|^2_{\ell^2}$. A centered placement $v \in \eta\mathbb{Z}^3$ satisfies $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2$. Translate the probe by v, reflect its source into the N_* cells, and periodize to obtain $\widehat{a}_c$. The reflected copies use the two-form orientation signs.

Let $D^{\mathrm{ch}}[B]$ be the chamber Dirac operator with bond phase B. Define the free one-cell ratio and its ordinary derivatives by

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[q(t/e)\widehat{a}_0] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}, \qquad c_m = N_0^{(m)}(0), \quad m = 2, 4. \tag{3}$$

The free mass is positive even when $\boldsymbol{\mu}_N$ is signed. Write $\mathbb{E}^{(0)}_{n_*}$ for the same-mass reference law conditioned on zero flux, and set

$$H_2(x; v) = x^2 - v, \qquad H_4(x; v) = x^4 - 6vx^2 + 3v^2, \qquad W_m(x; v, c) = e^{-m}H_m(x; v) - c. \tag{4}$$

Here and below e is the charge, so e^{-m} is its inverse power. With $X_c = \langle d\widehat{a}_c, F^{\mathrm{u}}\rangle$, put

$$\Theta^{(0)}_m = \left| \mathbb{E}^{(0)}_{n_*} \prod_{c\in\{0,\dots,2L-1\}^3} W_m(X_c; v_\eta, c_m) \right|^{1/N_*}. \tag{5}$$

Let $z_{r,n}$ be the positive zero-flux matter factor in (16). At continuum mass $\bar{m}$ define

$$b(f) = -6\,\mathrm{tr}\left[\left((\bar{m} + \gamma\cdot\partial)^{-1} i\gamma\cdot d^*\beta\right)^4\right],$$

where $\gamma_\nu = \mathrm{diag}(\sigma_\nu, -\sigma_\nu)$. The displayed fourth power is trace class and its trace is real by [26e, Theorem A].

Theorem A (Volume transfer and component cumulants)**.** Fix the data above and a charge interval $0 < e \le e_* \le 1/3$. For each charge let $N_{\mathrm{ref}}(e)$ be a finite nonnegative integer. Assume the following conditions for every charge in this interval and every $N \ge N_{\mathrm{ref}}(e)$.

(i) The scale and placement conditions are

$$v_K \ge 8, \qquad e\sqrt{h}\log h \le 1, \qquad 6N_*Mv_K^2\exp(-v_K)(v_K\exp(-v_K) + \tfrac{1}{2}) \le \pi^2,$$

and

$$D^{\mathrm{buf}}_\beta + 3\eta \le \ell_c, \qquad v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3 \tag{6}$$

for each centered placement used.

(ii) There are finite nonnegative constants C_{mat} and C_{ref}, independent of charge, mesh and placement, such that, at the same $q, \boldsymbol{\mu}_N$ on the two reference tori,

$$\begin{aligned} L^3\log z_{r,n_*} - \log z_{r,Ln_*} &\le C_{\mathrm{mat}}, \\ \Theta^{(0)}_4 \le C_{\mathrm{ref}}e^{\mathfrak{e}}, \qquad (1+|c_2|)\Theta^{(0)}_2 &\le C_{\mathrm{ref}}e^{\mathfrak{e}}, \\ |c_2| + |c_4|^{1/2} \le C_{\mathrm{ref}}, \qquad |c_4 - 3c_2^2 - b(f)| &\le C_{\mathrm{ref}}(e + \eta^{1/4}), \end{aligned} \tag{7}$$

where

$$\mathfrak{e} = \tfrac{1}{2}(1/4 - 9\varepsilon) - 12\varepsilon, \qquad a_L = \frac{L^3 - 1}{(L-1)^3}, \qquad \sigma = \mathfrak{e} - \frac{2a_L}{N_*} > 0.$$

There is $C_U < \infty$, depending only on these fixed data and constants, such that the following assertions hold uniformly in the stated charge interval.

For each $N \geq N_{\mathrm{ref}}(e)$ there are $u_2, u_4 \geq 0$, independent of the outer cutoff T, with

$$u_4 \leq C_U e^{\sigma}, \qquad (1+|c_2|)u_2 \leq C_U e^{\sigma}, \qquad |c_4 - 3c_2^2 - b(f)| \leq C_U(e^{\sigma} + \eta^{1/4}). \tag{8}$$

For every $T \geq T^*(e) + 1$ and every finite $Y \subset (R^*/L)\mathbb{Z}^3$ whose points are incongruent modulo $R\mathbb{Z}^3$,

$$\left| \mathbb{E}_{N,T} \prod_{y \in Y} W_m(F^{\sharp}(\tau_y f^{\eta}); v_{\eta}, c_m) \right| \leq u_m^{|Y|}, \qquad m = 2, 4. \tag{9}$$

Here $\tau_y f(x) = f(x-y)$ and $\mathbb{E}_{N,T}$ is expectation under $\mathbb{P}^{\mathrm{af}}_{n;q,\boldsymbol{\mu}_N}$. Its smooth-test fourth cumulant satisfies

$$|\kappa_4(F^{\sharp}(f)) - e^4 b(f)| \leq (8C_U + 3C_U^2)e^{4+\sigma} + C_U(\eta^2 + e^4\eta^{1/4}). \tag{10}$$

Fix one charge. Let μ be any prescribed volume-first limit: for $N_i \to \infty$, choose fixed-spacing weak volume limits μ_i of these laws, with $\mu_i \Rightarrow \mu$. All weak convergence here is on the negative Hermite space H_{-4} specified in section 7. Assume that μ is translation invariant and that almost every component in its translation-ergodic decomposition $\mu = \int \mu_\xi \, \mathrm{d}\nu(\xi)$ satisfies

$$\mathbb{E}_{\mu_\xi} \exp(F(g)) \leq \exp(\|g\|_2^2/2) \quad \text{for all real } g \in \mathcal{S}(\mathbb{R}^3; \Lambda^2). \tag{11}$$

Then, on one ν-full set,

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \leq C_f e^{4+\sigma}, \qquad C_f = 8C_U + 3C_U^2. \tag{12}$$

If $b(f) > 0$ and $C_f e^{\sigma} \leq b(f)/2$, every component in this set has $\kappa_4^{\mu_\xi}(f) \geq e^4 b(f)/2 > 0$. The full-measure set is for this fixed charge, probe and parent law. ■□

For the tuned chamber construction, the same mass assignment is selected on the top torus by [26a, Theorem A]. The reference-response estimate is [26c, Theorem A], and the two-torus matter comparison is [26h, Theorem A]. The free coefficients and a probe with $b(f) > 0$ are supplied by [26e, Theorems A–B]. The volume-first laws are constructed in [26d]; the component Gaussian bound is the last clause of [26b, Theorem A]. The simultaneous admission of the construction parameters is separate from the implication proved here.

The proof is the volume-transfer and component argument of [26g, §25–26], with its finite-periodic moment passage from [26g, §6.1–6.2]. Its first difficulty is normalization. Folding bounds the unnormalized observable by its value on the small reference torus, but the denominator belongs to the larger torus. Sidewise convexity of the partition function controls this denominator using the two reference volumes. The exact photon determinant then shows that the normalization costs only the power e^{-2a_L/N_*}; the response estimate retains the positive exponent σ.

The reference response is in flux zero. We split the full reference expectation by flux and bound its nonzero-flux part with the Gaussian tail and high moments. This gives bounds on every finite product of distinct translated Wick polynomials, uniformly in the outer volume. Their passage to a prescribed parent uses the periodic seminorm at finite volume and the sharp L^2 norm only after the supports fit in a period cube. Finally, spatial averages and a count of repeated indices bound the invariant conditional expectations of both Wick polynomials. These two conditional bounds determine the fourth cumulant in each component.

The charge restriction for positivity is imposed after C_U:

$$0 < e_\beta \leq \min\{e_*, [b(f)/(2C_f)]^{1/\sigma}\}. \tag{13}$$

It is positive when $b(f) > 0$ and $C_f > 0$. The conclusions concern subsequential limits of the stated cutoff family; no uniqueness of the continuum law is required.

2. The chamber law and the reference data

2.1. **Normalization, flux and Gaussian domination.** The chamber operator is the finite-range operator of [26f, §2]; its bond argument is a phase. It acts on four spin components and 48 chamber labels per site. The fermions are antiperiodic in each coordinate. For a rectangular coordinate torus with V sites and every side at least four, put $r_q = 2\pi/q$. The unnormalized integral of a curvature observable O is

$$Z[O] = \sum_{\substack{z\in C^2(\mathbb{Z})\\ dz=0}} \int_{[0,r_q)^{3V}} O(dA + r_q z)\exp\bigl(-\tfrac{1}{2}\|dA + r_q z\|^2\bigr)\det(D^{\mathrm{ch}}[qA] + \boldsymbol{\mu})\,\mathrm{d}A. \tag{14}$$

Cochains and the integer sum are on this torus; norms and pairings count positively oriented cells. The identification $(A, z) \sim (A + r_q k, z - dk)$ makes the integral independent of the chosen fundamental domain. The law is $Z[O]/Z[1]$. The partition function is finite and positive for every real $\boldsymbol{\mu}$, and the determinant gives nonnegative weights; see [26f, §2–3]. Flux is the three integer periods of z, ordered as $(23, 13, 12)$. We retain every flux sector in (14).

The coordinate-torus Gaussian bound [26f, Proposition 5.1] is

$$\mathbb{E}^{\mathrm{af}}\exp\langle g, F^{\mathrm{u}}\rangle \le \exp(\|g\|^2/2) \tag{15}$$

for every real torus two-cochain g. It holds for both odd and even coordinate sides. The law is invariant under lattice translations and $F^{\mathrm{u}} \mapsto -F^{\mathrm{u}}$. In particular all odd moments of a real linear observable vanish.

On $C(n)$, let γ_n be the standard Gaussian measure on the exact real two-cochains, and let $A(u)$ be the mean-zero coclosed potential of u. For $\theta \in [-\pi, \pi]^3$ put $H_{\theta,\nu} = \theta_\nu/(qn)$. The matter factor is

$$z_{r,n} = \int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int \det(D^{\mathrm{ch}}[q(A(u) + H_\theta)] + \boldsymbol{\mu})\,\mathrm{d}\gamma_n(u) > 0. \tag{16}$$

The subscript r denotes the regulator. The exact zero-flux factorization of [26c, §3.4] gives

$$Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n}, \tag{17}$$

$$Z_n^{\mathrm{ph}} = r_q^{n^3+2}(2\pi)^{n^3-1}\exp(-\mathcal{L}(n)/2). \tag{18}$$

where $\mathcal{L}(n)$ is the scalar-Laplacian logarithmic determinant in section 4. Thus the normalized zero-flux curvature law is the Gaussian law weighted by the twist-averaged determinant in (16). The factorization retains the gauge and harmonic volume factors; they will contribute to the partition defect.

2.2. **The source and cell contract.** The source identities and estimates in this subsection are proved in [26c, §2]. We state the needed contract with its support and normalization.

Write $p_{\mathrm{cell}} = n_*/L$ and $w = p_{\mathrm{cell}}/2$. The reference side n_* and p_{cell} are odd, so w is half-integral. The number $2L$ of cells per side is even. In unit-lattice coordinates the open cells are $Q_c^\circ = w(c + (0,1)^3)$.

For a two-form g let

$$g_{[\eta]}(\sigma) = \eta^{-3/2}\int_{Q_\eta(\sigma)} g_{\mu\nu}(y)\,\mathrm{d}y, \qquad Q_\eta(\sigma) = \eta(x + (e_\mu + e_\nu)/2 + [-1/2, 1/2)^3),$$

when $\sigma = (x; \mu\nu)$. The source of the introduction obeys $d\widehat{a}^o = (f^\eta)_{[\eta]}$. After periodizing, this gives

$$\langle d\widehat{a}, F^{\mathrm{u}}\rangle = F^\sharp(f^\eta).$$

All f^η are supported in Q_β for $\eta \le 1$, and

$$\|f^\eta - f\|_{H^s} \le \eta^2\|\beta\|_{H^{s+4}} \quad (s \in \mathbb{R}). \tag{19}$$

The variance converges with the quantitative estimate

$$\left|\sqrt{v_\eta} - \|f\|_2\right| \le 21\eta|\beta|_{C^3}(D_\beta^{\mathrm{buf}})^{3/2} + \eta^2\|\beta\|_{H^4}. \tag{20}$$

Here $|\beta|_{C^3}$ is the maximum component derivative norm through order three. These estimates follow from the centered difference symbols and the cell-average approximation.

For the placed probe, the coordinate map g_c carrying the base cell to cell c is, in coordinate μ, $x_\mu \mapsto x_\mu + 2iw$ if $c_\mu = 2i$, and $x_\mu \mapsto (2i+2)w - x_\mu$ if $c_\mu = 2i+1$. Transport oriented cochains by this map before periodizing. Denote the placed and reflected covering source by $\widehat{a}_c^{\,o}$; its periodization is $\widehat{a}_c$. On every carrying torus with $p_{\mathrm{cell}} \mid n$ and $n \ge n_*$,

$$\langle d\widehat{a}_c, d\widehat{a}_{c'}\rangle_n = v_\eta \mathbf{1}_{\{c=c'\}}. \tag{21}$$

The placement condition (6) ensures that their support boxes are strictly inside the open cells and that the boxes enlarged by one embed in the torus. Each polynomial $P(X_c)$ is consequently read by the same local scalar integrand on every such torus.

Here is the scalar observable class used in the transfer, with the conventions of [26f, §4.1, (46)–(50)]. For a coordinate box B, let $B^{(j)}$ be its contained oriented lattice j-cells. A scalar local integrand is a Borel gauge-invariant function G of the real bonds in $B^{(1)}$ such that, for some $C < \infty$ and $\epsilon_B > 0$,

$$|G(\widehat{A})| \le C \exp\left[(\tfrac{1}{2} - \epsilon_B) \sum_{\sigma \in B^{(2)}} |d\widehat{A}(\sigma)|^2\right].$$

Gauge invariance means $G(\widehat{A} - d\lambda) = G(\widehat{A})$. A local lift of (A, z) is a real bond cochain $\widehat{A}$ on B with $d\widehat{A} = F^{\mathrm{u}}$ and $\exp(iq\widehat{A}(b)) = \exp(iqA(b))$. The cited construction proves existence and independence of the lift for these evaluations.

For an open set U, let $\mathcal{A}(U)$ be the linear span of products $\Phi(d\widehat{A}) \prod_j G_j(\widehat{A}|_{B_j})$, where the boxes $B_j \subset U$ are pairwise disjoint and Φ is a bounded Borel function of finitely many curvature values in U. On a quotient with periods Γ the boxes must embed: each family $B_j + [-1,1]^3 + \gamma$ is disjoint, and all $B_j + \gamma$ for distinct (j, γ) are disjoint.

For the present probe take a supporting coordinate box $B_0 \Subset Q_0^\circ$ as in [26c, §2.2] and $B_c = g_c B_0$. The local expression for $P(X_c)$ is $P(\sum_{\sigma \subset B_c} d\widehat{a}_c^{\,o}(\sigma) d\widehat{A}(\sigma))$. Cauchy–Schwarz bounds its argument by $\sqrt{v_\eta}(\sum_{\sigma \subset B_c} |d\widehat{A}(\sigma)|^2)^{1/2}$, so it has a bound $C_P \exp(\frac{1}{4}\sum_{\sigma \subset B_c} |d\widehat{A}(\sigma)|^2)$. Thus it belongs to $\mathcal{A}(Q_c^\circ)$. The reflection is antilinear on coefficients; it fixes the coefficients of these real Wick polynomials.

Even cell indices give the precise translation formula

$$c(y) = \frac{2y}{p_{\mathrm{cell}}\eta} \in 2\mathbb{Z}^3, \qquad X_{c(y)} = F^\sharp(\tau_{v+y} f^\eta), \quad y \in (R^*/L)\mathbb{Z}^3. \tag{22}$$

Two such cells coincide on the torus exactly when the translations agree modulo its physical periods. Lattice translation invariance removes the common placement v from their product expectation.

2.3. **The normalized source identity.** On the reference torus define $a(t) = e^{-1}\sum_c t_c \widehat{a}_c$ and

$$\mathring{\Phi}(t) = z_{r,n_*}^{-1} \int \frac{\mathrm{d}\theta}{(2\pi)^3} \int \det(D^{\mathrm{ch}}[q(A(u) + H_\theta + a(t))] + \boldsymbol{\mu}_N)\, \mathrm{d}\gamma_{n_*}(u). \tag{23}$$

The θ integral is over $[-\pi, \pi]^3$. The source-shift identity of [26c, §3.6, Proposition 3.2 and (53)], valid for arbitrary real lattice mass, is

$$\mathbb{E}_{n_*}^{(0)} \prod_c W_m(X_c; v_\eta, c_m) = \left[\prod_c (\partial_{t_c}^m - c_m)\right] \mathring{\Phi}(0), \qquad m = 2, 4. \tag{24}$$

Thus (5) is exactly the response norm controlled there. Its reference law is normalized, and its coefficients are the ordinary free derivatives (3). The vacuum parameter of a common tuned trajectory cancels in this ratio. In a two-volume logarithmic defect its extensive vacuum contributions cancel as well.

3. Transfer from the reference volumes

The renormalization estimates are obtained on two reference tori. To use them for volume-first limits, we need bounds on products of local observables in arbitrarily large cutoff tori. The chessboard estimate reduces each such product to a folded one-cell observable. A separate convexity estimate controls the normalization by the partition functions of the two reference volumes.

3.1. **Folded observables and the partition defect.** Use unit-lattice lengths in this subsection. Fix the charge $q > 0$ and a real lattice mass. Choose coordinate cells of width $w \in \frac{1}{2}\mathbb{Z}_{>0}$ and offsets $o_\mu \in \frac{1}{2}\mathbb{Z}$:

$$Q_c^\circ = \{y : o_\mu + wc_\mu < y_\mu < o_\mu + w(c_\mu + 1),\ \mu = 1, 2, 3\}.$$

The walls are admissible coordinate mirrors. For even positive integers s_μ with $ws_\mu \geq 4$, let $C(ws)$ be the rectangular torus with these side lengths and antiperiodic fermions. It has $s_1 s_2 s_3$ cells. The wall-reflection group acts simply transitively on the covering cells. Write $O^{[c]}$ for the reflected copy of $O \in \mathcal{A}(Q_0^\circ)$ in Q_c°, and

$$O_{ws}^\sharp = \prod_{c \bmod s} O^{[c]}, \qquad \mathcal{Z}_s(O) = Z_{C(ws)}[O_{ws}^\sharp].$$

The copies use the antilinear reflection action, including conjugation of scalar coefficients.

These arrays satisfy the rectangular chessboard estimate [26f, (69)]. The cell theorem is stated there for integer width. For the coordinate cells used here, its fold proof also applies at half-integer width: all walls remain admissible, paired reflections translate by the integer $2w$, and the even cell counts give integer torus sides at least four. For each layer, the fold identity is applied with the trace based at that layer's own lower boundary. For a support box $B = y + [-r, r]^3 \subset Q_c^\circ$ one has $2r < w$. A nonzero period has sup norm at least $2w > 4r$ and at least four. Thus its enlarged boxes $B + [-1, 1]^3$ are disjoint, by the same embedding estimate used in the proof of [26f, Proposition 4.3]. The products therefore belong to the required slab classes.

Hold two entries of s fixed and vary s_μ. The product of the copies in the layer $c_\mu = 0$ gives a Hilbert–Schmidt slab operator Y independent of s_μ. Its reflected layer pairs fill the entire folded array. The slab-fold identity [26f, Proposition 4.2, (66)] gives

$$\mathcal{Z}_s(O) = \operatorname{Tr}(Y^*Y)^{s_\mu/2}. \tag{25}$$

In particular $\mathcal{Z}_s(O) \geq 0$, and $s_\mu \mapsto \mathcal{Z}_s(O)^{1/s_\mu}$ is nonincreasing. If $Y \neq 0$, the logarithm of this folded partition function is convex in s_μ. The first assertion is the monotonicity of Schatten norms, and the second follows from Hölder's inequality for the trace of positive powers. The normalization of Y here is that of the slab operators in [26f, §4.3].

Choose an even reference cell count $s_0 \geq 2$ with $n_r = ws_0 \geq 4$, and an integer $j \geq 2$. Put

$$N_r = s_0^3, \qquad D_j = j^3 \log Z_{C(n_r)}[1] - \log Z_{C(jn_r)}[1], \qquad \delta_j = \frac{D_j}{(j-1)^3 N_r}. \tag{26}$$

For an observable in the base cell its normalized reference seminorm is

$$\|O\|_* = \left(\frac{\mathcal{Z}_{(s_0,s_0,s_0)}(O)}{Z_{C(n_r)}[1]} \right)^{1/N_r}. \tag{27}$$

The partition function is positive, and the numerator is nonnegative, so this is well-defined also when it vanishes.

Proposition 3.1. One has $D_j \geq 0$. Let s be even, with either $s = s_0$ or $s \geq js_0$, and put $r = s/s_0$, $n = ws$. For every set $\mathcal{C}$ of distinct cells of $C(n)$ and $O_c \in \mathcal{A}(Q_0^\circ)$,

$$\left| \mathbb{E}_{C(n)} \prod_{c \in \mathcal{C}} O_c^{[c]} \right| \leq \prod_{c \in \mathcal{C}} \exp(\delta_j) \|O_c\|_*. \tag{28}$$

The right side uses only the two reference partition functions and the one-cell observables folded on $C(n_r)$. ∎

Proof. Successively decreasing the three side counts in (25) gives, for $s \ge s_0$,

$$\mathcal{Z}_{(s,s,s)}(O) \le \mathcal{Z}_{(s_0,s_0,s_0)}(O)^{r^3}. \tag{29}$$

Taking $O = 1$ and $r = j$ proves $D_j \ge 0$.

We next obtain the lower bound for the normalizing partition function. Define $p(r_1, r_2, r_3) = \log Z_{C(r_1 n_r, r_2 n_r, r_3 n_r)}[1]$ on ratios with integer side lengths at least four. The pressure trace [26f, (75)] shows that p is convex in each argument and that p/r_μ is nonincreasing in that argument. For $r \ge j$ put $a = (r-1)/(j-1)$. Convexity gives $p(r,\cdot,\cdot) \ge ap(j,\cdot,\cdot) - (a-1)p(1,\cdot,\cdot)$. Apply this in each argument, retaining the positive coefficient a at the next step. Bound the three subtracted terms by the side monotonicity. The result is

$$\begin{aligned} p(r,r,r) &\ge a^3 p(j,j,j) \\ &\quad - (a-1)(a^2j^2 + ajr + r^2)p(1,1,1) \\ &= r^3 p(1,1,1) - a^3 D_j. \end{aligned}$$

The last equality uses $aj - (a-1) = r$ three times. The negative coefficients are why upper bounds are used on the three subtracted terms; no sign condition on p is needed. Thus, including the equality case $r = 1$,

$$\log Z_{C(n)}[1] \ge r^3 \left(\log Z_{C(n_r)}[1] - \frac{D_j}{(j-1)^3} \right). \tag{30}$$

Fill the cells outside $\mathcal{C}$ with the constant integrand. The normalized rectangular chessboard estimate then gives

$$\left| \mathbb{E}_{C(n)} \prod_{c \in \mathcal{C}} O_c^{[c]} \right| \le \prod_{c \in \mathcal{C}} \left(\frac{\mathcal{Z}_{(s,s,s)}(O_c)}{Z_{C(n)}[1]} \right)^{1/(r^3 N_r)}.$$

Use (29) in the numerator and (30) in the denominator. Each factor is at most $\exp(\delta_j)\|O_c\|_*$, proving (28). □

4. The photon, matter and flux defect

4.1. **The scalar-Laplacian determinant.** The zero-flux Gaussian factor involves the determinant of the unit-lattice scalar Laplacian with its constant mode removed. Write its logarithm as

$$\mathcal{L}(n_1, n_2, n_3) = \sum_{\substack{k \in \prod_\mu \mathbb{Z}/n_\mu\mathbb{Z} \\ k \ne 0}} \log \left(\sum_{\mu=1}^{3} \left(2 - 2\cos \frac{2\pi k_\mu}{n_\mu} \right) \right), \qquad \mathcal{L}(n) = \mathcal{L}(n,n,n).$$

For $j \ge 2$ put

$$G_j = \left(1 + \frac{j}{3.2} \right) \frac{2j^2}{2.56}, \qquad K_j = (1 + j + j^2)(\log j + jG_j).$$

The estimate used for the two reference volumes is

$$\frac{1}{2}(\mathcal{L}(jn) - j^3 \mathcal{L}(n)) \le -(j^3 - 1)\log n + K_j \qquad (n, j \in \mathbb{Z},\ n, j \ge 2). \tag{31}$$

Its logarithmic term cancels the corresponding scale dependence of the Gaussian normalization. We retain the finite-volume calculation since a bulk determinant bound would lose this term.

Sum first in the first coordinate. For a nonzero transverse momentum $k' = (k_2, k_3)$ define $\omega(k') > 0$ by

$$2\cosh \omega(k') = 2 + \sum_{\mu=2,3} \left(2 - 2\cos \frac{2\pi k_\mu}{n_\mu} \right).$$

Factoring over the n_1-th roots of unity gives

$$\prod_{k_1\in\mathbb{Z}/n_1\mathbb{Z}}\left(2\cosh\omega-2\cos\frac{2\pi k_1}{n_1}\right)=4\sinh^2(n_1\omega/2).$$

For $k'=0$, omit $k_1=0$; the corresponding product is n_1^2. Therefore

$$\mathcal{L}(n_1,n_2,n_3)=2\log n_1+2\sum_{k'\neq0}\log\bigl(2\sinh(n_1\omega(k')/2)\bigr). \tag{32}$$

For $z>0$ let $g(z)=-\log(1-\exp(-2z))$. The identity $2\sinh z=\exp(z)(1-\exp(-2z))$ shows that

$$\log(2\sinh(jz))-j\log(2\sinh z)\le jg(z).$$

Consequently

$$\begin{aligned}\mathcal{L}(jn_1,n_2,n_3)-j\mathcal{L}(n_1,n_2,n_3)&\\ \le 2\log(jn_1)-2j\log n_1+2j\sum_{k'\neq0}g(n_1\omega(k')/2).&\end{aligned} \tag{33}$$

In the three applications below, $n_1=n$ and $n_2,n_3\in\{n,jn\}$. Represent each transverse momentum by $|k_\mu|\le n_\mu/2$ and put $p_\mu=2\pi k_\mu/n_\mu$. For $M(p)=\sum_{\mu=2,3}(2-2\cos p_\mu)$,

$$\frac{4}{\pi^2}|p|^2\le M(p)\le 8,\qquad \omega=2\,\mathrm{arcsinh}(\sqrt{M}/2)\ge 0.51|p|.$$

The last inequality follows from concavity of arcsinh on $[0,\sqrt{2}]$ and $\mathrm{arcsinh}\,\sqrt{2}>0.81\sqrt{2}$. Thus $n\omega(k')/2\ge c_j|k'|$, where $c_j=1.6/j$. Since $g(z)\le\exp(-2z)(1+(2z)^{-1})$, and $|k'|\ge(|k_2|+|k_3|)/\sqrt{2}$, one obtains

$$\begin{aligned}\sum_{k'\neq0}g(n\omega(k')/2)&\le\left(1+\frac{1}{2c_j}\right)\left[\left(\sum_{i\in\mathbb{Z}}\exp(-\sqrt{2}c_j|i|)\right)^2-1\right]\\ &=\left(1+\frac{1}{2c_j}\right)\sinh(c_j/\sqrt{2})^{-2}\le\left(1+\frac{1}{2c_j}\right)\frac{2}{c_j^2}=G_j.\end{aligned}$$

Enlargement to all of $\mathbb{Z}^2\setminus\{0\}$ is valid because the chosen representatives are distinct.

Finally telescope the enlargement of the three sides:

$$\begin{aligned}\mathcal{L}(jn,jn,jn)-j^3\mathcal{L}(n,n,n)=&\,[\mathcal{L}(jn,jn,jn)-j\mathcal{L}(n,jn,jn)]\\ &+j[\mathcal{L}(jn,n,jn)-j\mathcal{L}(n,n,jn)]\\ &+j^2[\mathcal{L}(jn,n,n)-j\mathcal{L}(n,n,n)].\end{aligned}$$

Symmetry of $\mathcal{L}$ supplies this identity. Each bracket is bounded by $2\log j+2(1-j)\log n+2jG_j$ by (33). Multiplication by $1,j,j^2$ and summation prove (31).

4.2. **The photon, matter and flux contributions.** Fix $q>0$ and a real lattice mass. On a cubic torus with $n\ge4$ sites per side, let $Z_n^{(w)}[1]$ be the partition function in flux sector $w\in\mathbb{Z}^3$, and put $P_n(w)=Z_n^{(w)}[1]/Z_n[1]$. The flux order is $(23,13,12)$. Define

$$\varsigma_n=\sum_{w\neq0}\exp\left(-\frac{2\pi^2|w|_2^2}{q^2n}\right). \tag{34}$$

The notation $z_{r,n}$ below is the positive matter factor of (16).

Proposition 4.1. For $n \geq 4$ and an integer $j \geq 2$,

$$Z_n[1] = \frac{Z_n^{\mathrm{ph}} z_{r,n}}{P_n(0)}, \qquad 0 < P_n(0) \leq 1, \tag{35}$$

$$P_n(w) \leq \exp\left(-\frac{2\pi^2 |w|_2^2}{q^2 n}\right), \qquad P_n(0) \geq 1 - \varsigma_n. \tag{36}$$

If $q^2 n \leq 1$, then

$$\varsigma_n \leq 7 \exp\left(-\frac{2\pi^2}{q^2 n}\right) < \frac{1}{8}. \tag{37}$$

The defect D_j of (26) decomposes as

$$D_j = D_j^{\mathrm{ph}} + D_j^{\mathrm{mat}} + D_j^{\mathrm{fl}}, \tag{38}$$

$$D_j^{\mathrm{mat}} = j^3 \log z_{r,n} - \log z_{r,jn}, \qquad D_j^{\mathrm{fl}} = -j^3 \log P_n(0) + \log P_{jn}(0),$$

$$\begin{aligned} D_j^{\mathrm{ph}} &= (j^3 - 1) \log \frac{2\pi}{q^2} + \tfrac{1}{2}[\mathcal{L}(jn) - j^3 \mathcal{L}(n)] \\ &\leq (j^3 - 1) \log \frac{2\pi}{q^2 n} + K_j. \end{aligned} \tag{39}$$

If $\varsigma_n < 1$, then

$$D_j^{\mathrm{fl}} \leq -j^3 \log(1 - \varsigma_n). \tag{40}$$

All identities use the same charge and lattice mass at the two sides; the mass can have either sign. ∎

Proof. The zero-flux factorization (17) gives $Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n} > 0$. All sectors have nonnegative weights, and their absolutely convergent sum is $Z_n[1]$. Hence $0 < P_n(0) \leq 1$ and (35).

For a configuration of flux w, let h_w be the constant two-cochain with value $r_q w_{\mu\nu}/n^2$ on plaquettes of orientation $\mu\nu$. The Fourier Hodge decomposition writes $r_q z = dB + h$ with h constant. Summing over a coordinate two-torus annihilates dB and gives $n^2 h_{\mu\nu} = r_q w_{\mu\nu}$. Thus $F - h_w$ is exact. A constant cochain is orthogonal to every exact cochain, by periodic summation by parts. In particular $\langle h, F \rangle = \langle h, h_w \rangle$ on sector w whenever h is constant.

Apply Gaussian domination (15) with $h = h_{w'}$ and split the expectation into its sectors:

$$1 \geq \sum_w \exp(\langle h_{w'}, h_w \rangle - \|h_{w'}\|^2/2) P_n(w) \geq \exp(\|h_{w'}\|^2/2) P_n(w').$$

Since $\|h_w\|^2 = r_q^2 |w|_2^2/n$, this proves the first bound in (36). The identity $\sum_w P_n(w) = 1$ proves the second.

Set $x = \exp(-2\pi^2/(q^2 n))$. Positivity permits the factorization

$$\varsigma_n = \left(\sum_{i \in \mathbb{Z}} x^{i^2}\right)^3 - 1 \leq \left(1 + \frac{2x}{1-x}\right)^3 - 1.$$

For $q^2 n \leq 1$, one has $x < 2^{-18}$. The bound $(1-x)^{-1} \leq 1 + 2x$ gives $2x/(1-x) \leq 2.001x < 10^{-5}$. Therefore $\varsigma_n \leq 2.001x(3 + 3 \cdot 10^{-5} + 10^{-10}) \leq 7x < 1/8$. This also proves convergence of (34).

Taking logarithms in (35) at n and jn gives (38). By (18),

$$\log Z_n^{\mathrm{ph}} = (n^3 + 2) \log r_q + (n^3 - 1) \log(2\pi) - \mathcal{L}(n)/2.$$

Its bulk terms cancel in the defect. The remaining constant is $(j^3 - 1) \log(2\pi/q^2)$. Equation (31) proves (39). Finally $\log P_{jn}(0) \leq 0$, and $P_n(0) \geq 1 - \varsigma_n > 0$ gives (40). □

5. Wick products on larger cutoff tori

Use the cell sources of Section 2.2 and write

$$N_* = 8L^3, \qquad a_L = \frac{L^3-1}{(L-1)^3}, \qquad D^{\mathrm{mat}} = L^3 \log z_{r,n_*} - \log z_{r,Ln_*}, \qquad \delta = \frac{D_L}{(L-1)^3 N_*}.$$

Fix $0 < e \le 1/3$, with $q = e\sqrt{\eta}$ and $q^2 n_* = e^2 R^*$, and assume

$$6N_* e^2 R^* \log(1/e) \le \pi^2. \tag{41}$$

The choice of scales ensuring this condition is given below. Let $\mathcal{C}_* = \{0, \dots, 2L-1\}^3$ and use the probe variance v_η of (21). For real c_2, c_4 put

$$O_m^{[c]} = W_m(X_c; v_\eta, c_m), \qquad m = 2, 4, \tag{42}$$

$$\Theta_m^{(0)} = \left| \left[\prod_{c \in \mathcal{C}_*} (\partial_{t_c}^m - c_m) \right] \mathring{\Phi}(0) \right|^{1/N_*}. \tag{43}$$

$$M_4 = 600 N_*^2 v_\eta^2 e^{-4} + |c_4|, \qquad M_2 = 12 N_* v_\eta e^{-2} + |c_2|,$$

$$T_m = \Theta_m^{(0)} + 2e^6 M_m, \qquad u_m = \exp(\delta) T_m. \tag{44}$$

Here $\mathring{\Phi}$ is the normalized external partition function (23). The notation e^{-m} denotes an inverse power of the charge.

Proposition 5.1. Assume the probe placement condition (6). For $n = n_*$, or $n \in p_{\mathrm{cell}}\mathbb{Z}$ with $n \ge Ln_*$, the all-flux law with the fixed real lattice mass obeys

$$0 \le \delta \le \frac{a_L}{N_*} \log \frac{2\pi}{e^2 R^*} + \frac{K_L + D^{\mathrm{mat}} + 1}{(L-1)^3 N_*}, \tag{45}$$

$$\left| \mathbb{E}_n^{\mathrm{af}} \prod_{c \in \mathcal{C}} O_m^{[c]} \right| \le u_m^{|\mathcal{C}|} \tag{46}$$

for every set $\mathcal{C}$ of distinct cells and $m = 2, 4$. In particular, if κ_j denotes the cumulant of X_0 on that torus, then

$$\left| e^{-2}(\kappa_2 - v_\eta) - c_2 \right| \le u_2, \tag{47}$$

$$\left| e^{-4}\kappa_4 - (c_4 - 3c_2^2) \right| \le u_4 + 3u_2(2|c_2| + u_2). \tag{48}$$

The bounds are uniform in n. Their right sides are formed on the two reference tori; $\Theta_m^{(0)}$, and therefore u_m, may depend on the placement. ∎

Proof. Condition (41) implies $e^2 R^* < 1$, since $N_* \ge 1000$ and $\log(1/e) \ge \log 3 > 1$. Put $x = \exp(-2\pi^2/(e^2 R^*))$. The same condition gives

$$x \le e^{12N_*}, \qquad x^{1/(2N_*)} \le e^6.$$

Apply Proposition 4.1 at n_* and $j = L$. Since $\varsigma_{n_*} \le 7x < 1/8$,

$$D_L^{\mathrm{fl}} \le -L^3 \log(1 - \varsigma_{n_*}) \le 8L^3 x = N_* x \le N_* 3^{-12N_*} \le 1.$$

The photon estimate and the defect decomposition now prove the upper bound in (45); its lower bound is Proposition 3.1.

All the polynomials in (42) are the same local integrands on every carrying torus, by Section 2.2. They are real elements of the cell observable class. Write $O_m^\sharp = \prod_{c \in \mathcal{C}_*} O_m^{[c]}$ on $C(n_*)$. Its all-flux expectation is nonnegative, by the fold trace (25). We estimate it by splitting flux zero from the other sectors.

First, Gaussian domination and the moment bound give, for every positive integer i,

$$\mathbb{E}^{\mathrm{af}}_{n_*} X_c^{2i} \le (2i\exp(1)v_\eta)^i.$$

Minkowski's inequality at exponent $2N_*$ yields

$$\|H_4(X_c;v_\eta)\|_{2N_*} \le [64\exp(2)N_*^2 + 24\exp(1)N_* + 3]v_\eta^2 \le 600N_*^2v_\eta^2,$$
$$\|H_2(X_c;v_\eta)\|_{2N_*} \le [4\exp(1)N_* + 1]v_\eta \le 12N_*v_\eta.$$

Thus $\|O_m^{[c]}\|_{2N_*} \le M_m$. The finite moments justify all uses of Hölder and Cauchy–Schwarz below, also when $v_\eta = 0$.

On the zero-flux sector, (24) gives

$$\left|\mathbb{E}^{\mathrm{af}}_{n_*}[O_m^\sharp \mathbf{1}_{\{\mathrm{flux}=0\}}]\right| = P_{n_*}(0)\left|\mathbb{E}^{(0)}_{n_*} O_m^\sharp\right| \le (\Theta_m^{(0)})^{N_*}.$$

On the remaining sectors, Cauchy–Schwarz followed by Hölder with N_* factors gives

$$\begin{aligned}\left|\mathbb{E}^{\mathrm{af}}_{n_*}[O_m^\sharp \mathbf{1}_{\{\mathrm{flux}\neq 0\}}]\right| &\le \varsigma_{n_*}^{1/2}\big(\mathbb{E}^{\mathrm{af}}_{n_*}|O_m^\sharp|^2\big)^{1/2}\\ &\le \varsigma_{n_*}^{1/2}\prod_{c\in\mathcal{C}_*}\|O_m^{[c]}\|_{2N_*}\\ &\le \big(\varsigma_{n_*}^{1/(2N_*)}M_m\big)^{N_*}.\end{aligned}$$

Since $\varsigma_{n_*}^{1/(2N_*)} \le 2e^6$, the nonnegative N_*-th root satisfies

$$\|O_m\|_* = \big(\mathbb{E}^{\mathrm{af}}_{n_*} O_m^\sharp\big)^{1/N_*} \le \Theta_m^{(0)} + 2e^6M_m = T_m.$$

For nonnegative a, b we used $(a^{N_*} + b^{N_*})^{1/N_*} \le a + b$. Proposition 3.1, with $s_0 = 2L$ and $j = L$, proves (46).

For one cell the same estimate gives $|e^{-m}\mathbb{E}H_m(X_0;v_\eta) - c_m| \le u_m$. Charge conjugation makes the odd moments zero. Writing $y = e^{-2}\mathbb{E}H_2(X_0;v_\eta)$, one has

$$e^{-4}\kappa_4 = e^{-4}\mathbb{E}H_4(X_0;v_\eta) - 3y^2.$$

The second-order conclusion follows at once. For the fourth, use $|y - c_2| \le u_2$ and $|y + c_2| \le 2|c_2| + u_2$ to obtain (48). □

6. Reference scales and the loss of charge exponent

The reference side is chosen from the logarithmic charge scale. Use L, M, h from the introduction and the localization power two. Thus

$$e_K = e\sqrt{h}, \qquad v_K = -\log e_K, \qquad r_K = \min\{L^i : L^i \ge v_K^2,\ i \in \mathbb{Z}_{\ge 0}\}, \qquad R^* = \frac{hMr_K}{L}. \tag{49}$$

Assume $v_K \ge 8$, $e \le 1/3$, $e_K \log h \le 1$, and

$$6N_*Mv_K^2\exp(-v_K)\left(v_K\exp(-v_K) + \tfrac{1}{2}\right) \le \pi^2. \tag{50}$$

These are finite charge conditions after h, L, M are fixed. They hold on a sufficiently small charge interval: $v_K \to \infty$ as $e \to 0$, and the polynomial times exponential in (50) tends to zero.

Minimality gives $v_K^2 \le r_K < Lv_K^2$. Consequently $T^* = \log_L(Mr_K) - 1$ is an integer at least one, $R^* = hL^{T^*}$, and $\ell_c = R^*/(2L) \to \infty$ as $e \to 0$. Moreover $1/L < \bar{m}h \le 1$, so

$$\bar{m}R^* > \frac{Mr_K}{L^2} \ge r_K \ge v_K, \qquad \exp(-\bar{m}R^*) \le e\sqrt{h}. \tag{51}$$

The same upper bound on r_K gives

$$\begin{aligned}e^2R^*\log(1/e) &< Mv_K^2\exp(-2v_K)(v_K + \tfrac{1}{2}\log h)\\ &\le Mv_K^2\exp(-v_K)(v_K\exp(-v_K) + \tfrac{1}{2}).\end{aligned}$$

Thus (50) implies (41). The probe fits every sufficiently small charge and sufficiently fine mesh, since D_β^{buf} is fixed and $\ell_c \to \infty$.

We now isolate the two reference estimates used in the passage to arbitrary volumes. Their dependence on the charge matters: a bounded partition defect still loses a small power of e through its photon contribution. Let C_{mat} and C_{ref} be finite constants independent of the charge, mesh and placement. Assume that for $N \ge N_{\mathrm{ref}}(e)$ the mass assignment and reference response satisfy

$$\begin{aligned} D^{\mathrm{mat}} &\le C_{\mathrm{mat}}, \\ \Theta_4^{(0)} \le C_{\mathrm{ref}} e^{\mathfrak{e}}, &\qquad (1+|c_2|)\Theta_2^{(0)} \le C_{\mathrm{ref}} e^{\mathfrak{e}}, \end{aligned} \tag{52}$$

where

$$\mathfrak{e} = \tfrac{1}{2}(1/4 - 9\varepsilon) - 12\varepsilon, \qquad \sigma = \mathfrak{e} - \frac{2a_L}{N_*} > 0, \qquad 0 < \varepsilon \le 1/200. \tag{53}$$

Here $N_{\mathrm{ref}}(e)$ is a fixed finite mesh threshold. These are the reference hypotheses of theorem A. For the common tuned mass, the matter estimate is [26h, Theorem A, (5)], and the response estimate is [26c, Theorem A]. Their joint charge interval and cutoff threshold are part of the construction data assumed here.

On the reference torus take the free coefficients $c_m = N_0^{(m)}(0)$ from (3), using the source associated with the same placed probe. [26e, Proposition 5.2] and the support fit give a constant, enlarged into C_{ref} if necessary, such that

$$|c_2| + |c_4|^{1/2} \le C_{\mathrm{ref}}, \qquad |c_4 - 3c_2^2 - b(f)| \le C_{\mathrm{ref}}(e + \eta^{1/4}). \tag{54}$$

Indeed $R^* \ge D_\beta + 1$, and (51) bounds the finite-volume error in [26e, Theorem B]. The coefficients are real and independent of placement.

Proposition 6.1. Fix the probe and the scale data above, and assume (6), (52) and (54). There is a finite constant C_U, independent of e, N, T and placement, such that for all the stated reference charges and $N \ge N_{\mathrm{ref}}(e)$,

$$u_4 \le C_U e^{\sigma}, \qquad (1+|c_2|)u_2 \le C_U e^{\sigma}, \qquad |c_4 - 3c_2^2 - b(f)| \le C_U(e^{\sigma} + \eta^{1/4}). \tag{55}$$

On every cutoff torus with $T \ge T^* + 1$, every finite set $Y \subset (R^*/L)\mathbb{Z}^3$ of points incongruent modulo $R\mathbb{Z}^3$ satisfies

$$\left| \mathbb{E}_{N,T} \prod_{y \in Y} W_m(F^\sharp(\tau_y f^\eta); v_\eta, c_m) \right| \le u_m^{|Y|}, \qquad m = 2, 4, \tag{56}$$

Here W_m is the polynomial in (4). The fourth cumulant of the fixed smooth test obeys

$$|\kappa_4(F^\sharp(f)) - e^4 b(f)| \le (8C_U + 3C_U^2)e^{4+\sigma} + C_U(\eta^2 + e^4\eta^{1/4}). \tag{57}$$

The tests f^η have one fixed compact support, $f^\eta \to f$ in L^2, and $v_\eta \to \|f\|_2^2$. Thus (55) and (56) give exactly the cutoff hypotheses of Proposition 8.2. ∎

Proof. Write

$$s_\beta = \|f\|_2 + \|\beta\|_{H^4}, \qquad C_\Gamma^{\mathrm{tr}} = (2\pi)^{a_L/N_*} \exp\left(\frac{K_L + C_{\mathrm{mat}} + 1}{(L-1)^3 N_*} \right).$$

Since $R^* \ge 1$, (45) gives

$$\exp(\delta) \le C_\Gamma^{\mathrm{tr}} e^{-2a_L/N_*}. \tag{58}$$

The inverse power here is a power of the charge. The source-curvature identity and Jensen's inequality for the disjoint averaging cubes give $v_\eta^{1/2} \le \|f^\eta\|_2 \le s_\beta$. Use this and (54) in (44):

$$\begin{aligned} T_4 &\le C_{\mathrm{ref}} e^{\mathfrak{e}} + 1200 N_*^2 s_\beta^4 e^2 + 2C_{\mathrm{ref}}^2 e^6 \\ &\le (C_{\mathrm{ref}} + 1200 N_*^2 s_\beta^4 + 2C_{\mathrm{ref}}^2) e^{\mathfrak{e}}, \end{aligned}$$

$$(1+|c_2|)T_2 \le C_{\mathrm{ref}}e^{\mathfrak{e}} + 2e^6(12N_*s_\beta^2e^{-2} + C_{\mathrm{ref}})(1+C_{\mathrm{ref}})$$
$$\le [C_{\mathrm{ref}} + 2(12N_*s_\beta^2 + C_{\mathrm{ref}})(1+C_{\mathrm{ref}})]e^{\mathfrak{e}}.$$

Here $0 < \mathfrak{e} < 1/4$, from (53). One possible constant is

$$C_U = \max\{C_\Gamma^{\mathrm{tr}}(C_{\mathrm{ref}} + 1200N_*^2s_\beta^4 + 2C_{\mathrm{ref}}^2),$$
$$C_\Gamma^{\mathrm{tr}}[C_{\mathrm{ref}} + 2(12N_*s_\beta^2 + C_{\mathrm{ref}})(1+C_{\mathrm{ref}})],\ C_{\mathrm{ref}},\ 60s_\beta^4\}. \tag{59}$$

Multiplication by (58) proves the first two bounds in (55). The last follows from (54), since $e \le e^\sigma$.

For a cutoff side $n = 8hL^{T+N}$ with $T \ge T^* + 1$, one has $n = 8L^{T-T^*}n_*$, a multiple of p_{cell} with $n \ge Ln_*$. For $y \in (R^*/L)\mathbb{Z}^3$ use the even cell index $c(y)$ of (22). Incongruent points give distinct cells. Proposition 5.1 applies to their product, and lattice translation invariance removes the common placement. This proves (56). For a fixed finite set of distinct points it applies as soon as $R > 2\max_{y\in Y}|y|_\infty$.

It remains to replace the discrete smooth test f^η by f in the cumulant. We give the quantitative multilinear estimate used for this purpose. For a real two-cochain g, charge conjugation and Gaussian domination give

$$\mathbb{E}\cosh(t\langle g, F^{\mathrm{u}}\rangle) \le \exp(t^2\|g\|^2/2).$$

Extract its nonnegative term of degree $2i$ and choose $t^2 = 2i/\|g\|^2$. This gives

$$\mathbb{E}\langle g, F^{\mathrm{u}}\rangle^{2i} \le (2i)!\left(\frac{\|g\|^2}{2i}\right)^i \exp(i).$$

For $i = 2$ the bound is $(3/2)\exp(2)\|g\|^4$. Hölder bounds the four-field moment by

$$\tfrac{3}{2}\exp(2)\prod_a \|g_a\|.$$

The variance bound bounds each of its three pair subtractions by $\prod_a\|g_a\|$. Since $(3/2)\exp(2)+3 < 15$, the symmetric multilinear forms K_2, K_4 obey

$$|K_2(g_1,g_2)| \le \|g_1\|\|g_2\|, \qquad |K_4(g_1,g_2,g_3,g_4)| \le 15\prod_a\|g_a\|. \tag{60}$$

For $g = 0$ the corresponding moment is zero.

Let g and g' be the periodizations of the normalized covering cochains $f_{[\eta]}$ and $(f^\eta)_{[\eta]}$ from section 2.2 onto the cutoff torus. The common support cube has side less than R. The averaging bound and (19) give

$$\|g\|, \|g'\| \le s_\beta, \qquad \|g - g'\| \le \eta^2\|\beta\|_{H^4}.$$

The observables paired with these cochains are $F^\sharp(f)$ and $F^\sharp(f^\eta)$. Telescope the four slots of K_4 and the two slots of K_2 in (60). This proves

$$|\kappa_4(F^\sharp(f)) - \kappa_4(F^\sharp(f^\eta))| \le 60s_\beta^4\eta^2,$$
$$|\kappa_2(F^\sharp(f)) - \kappa_2(F^\sharp(f^\eta))| \le 2s_\beta^2\eta^2. \tag{61}$$

One-cell transfer and (55) bound the fourth cumulant of $F^\sharp(f^\eta)$ by

$$|\kappa_4(F^\sharp(f^\eta)) - e^4(c_4 - 3c_2^2)| \le (7C_U + 3C_U^2)e^{4+\sigma}.$$

Adding the free-coefficient error and the first bound in (61) proves (57). Finally the compact support, L^2 convergence and variance convergence are respectively (2), (19) and (20). □

7. Periodic fields and a prescribed continuum law

A finite periodic step field does not satisfy the sharp global $L^2(\mathbb{R}^3)$ Gaussian bound for arbitrary Schwartz tests. The periodized norm must be retained until the volume tends to infinity. We give the moment argument of [26g, §6.1–6.2]; it is the step needed to use the finite-volume product estimates for a prescribed parent.

Let $\mathsf{A} = 1 + |y|^2 - \Delta$ on $\mathbb{R}^3$, with real L^2-orthonormal Hermite two-forms h_κ and eigenvalues λ_κ. Each scalar Hermite function occurs once for each of the three components. We use the Hilbert space

$$H_{-4} = \left\{ F : \sum_\kappa \lambda_\kappa^{-8} |F(h_\kappa)|^2 < \infty \right\}.$$

It is continuously embedded in tempered two-form distributions, and each Schwartz evaluation is continuous there. The step-field realization, tightness and fixed-spacing limits in this space are proved in [26d].

7.1. **The periodic seminorm.** For $R = n\eta$ and a Schwartz two-form g, set $g^R(y) = \sum_{k\in\mathbb{Z}^3} g(y+Rk)$ and

$$g_\eta([\sigma]) = \eta^{-3} \int_{Q_\eta(\sigma)} g^R_{\mu\nu}(y)\,\mathrm{d}y, \qquad |g_\eta|_\eta^2 = \eta^3 \sum_{[\sigma]} |g_\eta([\sigma])|^2.$$

Unfolding the absolutely convergent sum gives

$$F^\sharp(g) = \eta^{3/2} \sum_{[\sigma]} g_\eta([\sigma]) F^{\mathrm{u}}(\sigma). \tag{62}$$

Cell Cauchy–Schwarz and Minkowski give

$$|g_\eta|_\eta \le \|g^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} \le p_R(g), \qquad p_R(g) = \|g\|_2 + 26R^{-2}\||y|^2 g\|_{L^2(\mathbb{R}^3\setminus Q_R)}, \tag{63}$$

where $Q_R = [-R/2, R/2)^3$. For completeness, put $a_k = \|g\|_{L^2(Q_R+Rk)}$. The periodized norm is at most $\sum_k a_k$. On a cube with $k \neq 0$, $|y| \ge R|k|_\infty/2$, so

$$\sum_{k\neq 0} a_k \le 4R^{-2} \left(\sum_{k\neq 0} |k|_\infty^{-4} \right)^{1/2} \||y|^2 g\|_{L^2(\mathbb{R}^3\setminus Q_R)}.$$

The shell count $24j^2 + 2$ gives a sum $4\pi^2 + \pi^4/45 < 41.65$; its square root times four is less than 26. This proves (63). For $R \ge 1$,

$$p_R(g) \le p_1(g), \qquad p_R(g) \longrightarrow \|g\|_2 \quad (R \longrightarrow \infty).$$

The seminorm p_1 is continuous on Schwartz space. If the support of g lies in a translate of an open R-cube, its distinct period translates are disjoint and $|g_\eta|_\eta \le \|g\|_2$.

By (15),

$$\mathbb{E}_{N,T} \exp(F^\sharp(g)) \le \exp(|g_\eta|_\eta^2/2) \le \exp(p_R(g)^2/2) \qquad (g \text{ real}). \tag{64}$$

Proposition 7.1. Suppose $N_i, T_i \to \infty$ and the laws P_{N_i,T_i} of $F^\sharp$ converge weakly on H_{-4} to μ. Then $\mathbb{E}_\mu \exp(F(g)) \le \exp(\|g\|_2^2/2)$ for every real Schwartz two-form g. Every joint moment of a fixed finite family of Schwartz tests converges along the same sequence. ∎

Proof. Truncate $\exp(F(g))$ above, pass this bounded continuous function to the limit, and remove the truncation. Since $R_i \to \infty$, (63) proves the Gaussian bound.

For fixed complex tests $g_1, \dots, g_s$ put

$$Z_i(z) = \mathbb{E}_{N_i,T_i} \exp\left(\sum_a z_a F^\sharp(g_a) \right), \qquad Z(z) = \mathbb{E}_\mu \exp\left(\sum_a z_a F(g_a) \right).$$

The evaluations are continuous on H_{-4}, and

$$\mathbb{E}_{N_i,T_i}\left|\exp\left(\sum_a z_a F^\sharp(g_a)\right)\right|^2 \le \exp\left(2p_1\left(\operatorname{Re}\sum_a z_a g_a\right)^2\right).$$

This is uniform in i and on compact z-sets. Uniform integrability and weak convergence give pointwise convergence of Z_i to Z. The exponential bounds make these functions entire and locally uniformly bounded. Cauchy's estimates give equicontinuity on compact sets; pointwise convergence is therefore locally uniform. Cauchy's formula passes every derivative at zero to the limit, which proves joint moment convergence. □

We also record the varying-test estimate used below. If $g_1,\dots,g_J$ are real and their supports lie in one open period cube, the Gaussian moment bounds and Hölder give

$$\mathbb{E}\prod_{a=1}^{J}|F^\sharp(g_a)| \le c_J\prod_{a=1}^{J}\|g_a\|_2$$

for a constant c_J depending only on J. The same estimate holds for μ. Thus replacing each factor from a bounded family of tests by another test, with L^2 difference at most δ, changes the expectation of a degree-J monomial by at most $Jc_JB^{J-1}\delta$, where B bounds their L^2 norms. This follows by telescoping the J slots. For any fixed finite family of compactly supported translates, the support condition holds once R_i is sufficiently large.

7.2. **Realizing the chosen parent by finite tori.** Let μ_i and μ be the prescribed laws in theorem A. Fix a metric d_{w} for weak convergence of probability measures on the separable Hilbert space H_{-4}. At spacing $\eta_i = L^{-N_i}$, take the defining volume sequence for μ_i. Choose an element T_i of that sequence so large that

$$d_{\mathrm{w}}(P_{N_i,T_i},\mu_i) < 1/i, \qquad T_i \ge i + T^*(e) + 1.$$

Then

$$d_{\mathrm{w}}(P_{N_i,T_i},\mu) \le 1/i + d_{\mathrm{w}}(\mu_i,\mu) \longrightarrow 0.$$

Both cutoffs diverge. This is the prescribed-parent diagonal of [26d, Corollary 7.4], with the finite-periodic moment conclusion supplied by proposition 7.1. Discard finitely many terms so that $N_i \ge N_{\mathrm{ref}}(e)$. At this fixed charge the translation spacing $\lambda = R^*(e)/L$ is independent of i.

The four reference quantities $c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$ are bounded by the estimates of proposition 6.1. A subsequence on which they converge still has weak limit μ. For every fixed finite $Y \subset \lambda\mathbb{Z}^3$, the points are eventually incongruent modulo the torus and all the translated supports fit in a single open period cube. The next section passes the Wick products along this same coefficient subsequence. The threshold for support containment may depend on Y; the coefficients and parent law do not.

8. Fourth cumulants of the limiting components

The fourth-cumulant estimate for a mixed limiting law does not by itself control each ergodic component. We use the stronger estimate on products of translated Wick polynomials. It bounds their translation-invariant conditional expectations, and hence the second and fourth moments of almost every component.

8.1. **From distinct-point products to the invariant part.** Let $U(y)$, $y \in \mathbb{Z}^d$, be the operators of a measure-preserving action on a probability space, where $d \ge 1$, with $U(y)G = G \circ T_{-y}$. Let P_{lat} be the orthogonal projection in L^2 onto the vectors invariant under this action.

Lemma 8.1. Suppose that G is real and belongs to every L^p, $p < \infty$. If $u \geq 0$ and

$$\left| \mathbb{E} \prod_{a=1}^{J} U(y_a) G \right| \leq u^J \tag{65}$$

for every finite family of pairwise distinct points of $\mathbb{Z}^d$, then $|P_{\mathrm{lat}} G| \leq u$ almost surely. ■

Proof. For $B_n = \{0, \dots, n-1\}^d$ put $A_n = n^{-d} \sum_{y \in B_n} U(y)$. These contractions converge strongly in L^2 to P_{lat}. Indeed, the orthogonal complement of the invariant space is the closed span of $(U(y) - 1)H$. On such a vector, cancellation of the common terms in B_n and $y + B_n$ gives

$$\|A_n (U(y) - 1) H\|_2 \leq 2|y|_1 n^{-1} \|H\|_2.$$

On invariant vectors the averages are the identity, proving the asserted convergence by the orthogonal decomposition.

For an even integer $J \geq 2$, expand $\mathbb{E}(A_n G)^J$. The distinct-index terms obey (65). There are at most $\binom{J}{2} n^{d(J-1)}$ remaining tuples: each has a repeated pair of indices. Hölder's inequality bounds their individual expectations in modulus by $\|G\|_J^J$. Therefore

$$\mathbb{E}|A_n G|^J \leq u^J + \binom{J}{2} n^{-d} \|G\|_J^J.$$

Along a subsequence the L^2 convergence is almost sure. Fatou's lemma gives $\mathbb{E}|P_{\mathrm{lat}} G|^J \leq u^J$ for every even J. For $t > u$, Markov's inequality now gives $\Pr(|P_{\mathrm{lat}} G| > t) \leq (u/t)^J$. Let $J \to \infty$ and then $t \downarrow u$ through a countable sequence. □

Let a translation-invariant law μ on tempered fields have the decomposition $\mu = \int \mu_\xi \, \mathrm{d}\nu(\xi)$. Write I for its translation-invariant sigma algebra and $P = \mathbb{E}_\mu[\,\cdot \mid I]$. Apply the lemma to translations on $\lambda\mathbb{Z}^3$, with $\lambda > 0$ fixed. The range of P is contained in that of P_{lat}, so $P = P P_{\mathrm{lat}}$. Positivity of conditional expectation therefore gives

$$|PG| \leq P|P_{\mathrm{lat}} G| \leq u, \qquad |\mu_\xi(G)| \leq u \quad \text{for } \nu\text{-almost every } \xi. \tag{66}$$

The second assertion uses the conditional-law representation of the decomposition. Integrability under the conditional laws and the representation for this integrable G follow by truncating its positive and negative parts.

8.2. **Passing the Wick-product estimate to a limit law.** Fix the charge $e > 0$ and a real compactly supported test f. Use the Wick polynomials W_m in (4), with variance argument $v \geq 0$ and $m = 2, 4$.

Take a joint cutoff sequence defining a limit law μ. Suppose that the lattice estimates give compactly supported real tests f_i, all supported in one fixed cube, and real numbers

$$v_i, \; c_{2,i}, \; c_{4,i}, \; u_{2,i}, \; u_{4,i},$$

with $v_i \geq 0$, such that

$$\|f_i - f\|_2 \to 0, \qquad v_i \to \|f\|_2^2, \qquad \sup_i (|c_{2,i}| + |c_{4,i}|) < \infty. \tag{67}$$

For a fixed $\lambda > 0$, assume that

$$\left| \mathbb{E}_i \prod_{y \in Y} W_m(F^\sharp(\tau_y f_i); v_i, c_{m,i}) \right| \leq u_{m,i}^{|Y|}, \qquad m = 2, 4, \tag{68}$$

for every finite $Y \subset \lambda\mathbb{Z}^3$ and all sufficiently large i depending on Y. This is the translated-cell form of the transfer estimate; the volume condition makes the points of Y incongruent modulo the torus. The reference estimates also give

$$\begin{aligned} 0 \le u_{4,i} \le Ce^{\sigma}, \qquad 0 \le (1+|c_{2,i}|)u_{2,i} \le Ce^{\sigma}, \\ |c_{4,i} - 3c_{2,i}^2 - b(f)| \le C(e^{\sigma} + \eta_i^{1/4}), \end{aligned} \tag{69}$$

with C independent of the cutoff, $\sigma > 0$ and $0 < e \le 1$. For the chamber laws, Proposition 6.1 supplies these hypotheses throughout the charge interval (13).

Pass to a subsequence on which the four bounded sequences $c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$ converge, with limits c_2, c_4, u_2, u_4. The weak limit remains μ. Equation (69) gives

$$u_4 \le Ce^{\sigma}, \quad (1+|c_2|)u_2 \le Ce^{\sigma}, \quad |c_4 - 3c_2^2 - b(f)| \le Ce^{\sigma}. \tag{70}$$

Set $G_m = W_m(F(f); \|f\|_2^2, c_m)$. Then (68) passes to

$$\left| \mathbb{E}_{\mu} \prod_{y \in Y} U(y) G_m \right| \le u_m^{|Y|} \qquad (Y \subset \lambda\mathbb{Z}^3 \text{ finite}). \tag{71}$$

To justify this passage for the varying tests, expand the finite product of Wick polynomials into monomials. For a monomial of positive total degree J, telescope the replacement of each $F^{\sharp}(\tau_y f_i)$ by $F^{\sharp}(\tau_y f)$. All these tests have support in one open R_i-cube once i is large. The test-map variance and product-moment bounds then give an error at most

$$J c_J B^{J-1} \|f_i - f\|_2, \qquad B = \sup_i \max(\|f_i\|_2, \|f\|_2).$$

This tends to zero. The moments of the fixed tests converge by Section 7. The polynomial coefficients converge by the selected coefficient subsequence and the variance convergence in (67). There are only finitely many monomials for each fixed Y. This proves (71).

8.3. **The cumulant in an ergodic component.**

Proposition 8.2. Under (67)–(69), suppose that almost every translation-ergodic component of μ satisfies Gaussian domination. Then

$$\left| \kappa_4^{\mu_\xi}(f) - e^4 b(f) \right| \le (8C + 3C^2) e^{4+\sigma} \tag{72}$$

for almost every component. In particular, if $b(f) > 0$ and $(8C + 3C^2)e^{\sigma} \le b(f)/2$, its fourth cumulant is positive. ∎

Proof. Gaussian domination of μ makes the real polynomials G_2, G_4 belong to every $L^p(\mu)$. Apply Lemma 8.1 to (71), and then use (66). On one set of full component measure,

$$|d_m| \le u_m, \qquad d_m = \mathbb{E}_{\mu_\xi} G_m, \qquad m = 2, 4.$$

Intersect with the full-measure set of components having Gaussian domination. For a component in this intersection, put $X = F(f)$ and $v = \|f\|_2^2$. Gaussian domination gives $\mathbb{E}_{\mu_\xi} X = 0$ and all moments finite. Thus

$$\kappa_4^{\mu_\xi}(f) = \mathbb{E}_{\mu_\xi} H_4(X; v) - 3\big(\mathbb{E}_{\mu_\xi} H_2(X; v)\big)^2.$$

The definition of G_m gives the exact identity

$$e^{-4} \kappa_4^{\mu_\xi}(f) = c_4 + d_4 - 3(c_2 + d_2)^2. \tag{73}$$

Consequently (70) implies

$$\begin{aligned} |e^{-4}\kappa_4^{\mu_\xi}(f) - b(f)| &\le |c_4 - 3c_2^2 - b(f)| + u_4 + 6|c_2|u_2 + 3u_2^2 \\ &\le 8Ce^{\sigma} + 3C^2 e^{2\sigma} \le (8C + 3C^2)e^{\sigma}. \end{aligned}$$

This proves the assertion uniformly over the full-measure set of components. A centered normal variable has zero fourth cumulant, so positivity also proves that these laws are non-Gaussian. □

8.4. **Completion for the prescribed volume-first law.**

Proof of Theorem A. The scale calculation in section 6 gives (41) and the reference volume error. The reference and free-coefficient hypotheses are exactly (52) and (54). Thus proposition 6.1, with the uniform constant (59), proves (8)–(10).

For the prescribed parent, use the finite-torus diagonal in section 7. Set $f_i = f^{\eta_i}$ and $v_i = v_{\eta_i}$. The compact support, test convergence and variance convergence are (2), (19) and (20). At this fixed charge, $\lambda = R^*/L$ is fixed. Each finite $Y \subset \lambda\mathbb{Z}^3$ eventually consists of distinct torus translates. The four reference sequences have a convergent subsequence, whose weak limit is still the prescribed μ. The product passage proved above therefore applies to this same law, with $C = C_U$. Its components have Gaussian domination by the explicit hypothesis (11). Apply proposition 8.2 and impose (13) when $b(f) > 0$.

If a separate argument gives other properties on a full set of components of this parent, intersect that set with the one obtained here. The component laws themselves have not been replaced. □

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